Sleep Detection from Video

Take-home exercise, Computer Vision & AI internship, KRP Solutions · Abhinand

1. Problem and definition

Given an ordinary video of one or two people, the system says for every second whether each person is **awake, drowsy, asleep, unknown** (present but not visible enough) or **absent**, lists the asleep intervals, and names each person from one reference photo. Sleep cannot be measured from video, so I use an **observable, behavioural** definition that changes with what the camera can see:

What is visible	Asleep when…	Confidence
Face, eyes measurable	eyes closed (EAR below 0.5 × own open-eye level) and body still for **10 s**	0.9
Body only (face hidden, facing away)	body still for **45 s** in a sleep posture (slumped, head on arms, lying)	0.65
Mostly not visible	never guessed: reported as *unknown*	–

Closures longer than a 0.5 s blink but under 10 s, a high PERCLOS (eyes closed ≥ 30% of a 30 s window) or head nodding count as **drowsy**. Waking needs 2 s of open eyes or clear movement, so one noisy frame cannot flip the state.

2. Approach: pretrained models plus explainable rules

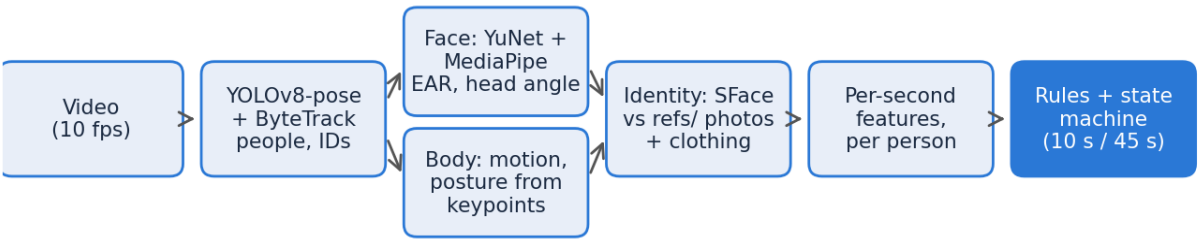


Figure 1. Pipeline. Nothing is trained; every decision is a rule with a threshold in *config.yaml*.

- **People:** YOLOv8n-pose finds each person and 17 body keypoints; ByteTrack keeps an ID across frames.
- **Eyes:** YuNet finds the face, MediaPipe Face Landmarker gives 478 points and head angles. The **Eye Aspect Ratio** (EAR) uses 6 points per eye: about 0.30 open, 0.05 closed. Each person's open level is calibrated on their first 10 s facing the camera. EAR is only trusted when the head is roughly frontal (turned < 35°, tilted down < 20° from the camera), because looking down makes eyes *look* closed.
- **Body:** motion energy = share of pixels changing inside the person's box, minus background change; posture from keypoints (head height above the shoulders, head near the arms, torso angle).
- **Identity:** SFace turns each face into 128 numbers, compared with *refs/<Name>.jpg*. Votes are collected over a whole track, so the name stays when the person turns away; one name can never be on two people at once; clothing colour is a backup after an ID switch.
- **Engineering:** the slow model pass is cached, so the rules re-run in seconds for tuning; 47 unit tests; Dockerfile built and tested; one command: *python run.py --video clip.mp4 --refs refs/*.

3. Data and evaluation

I recorded **9 clips (about 25 min)** on a laptop webcam: normal work, three 5 s eye closures, acted sleep facing the camera (×2), head down on the desk, sitting sideways on the phone, looking down at a phone, dim light (16 fps), and one clip with two people (Abhinand and Rajni, 3.5 min). Every second was labelled by hand. Only two thresholds were tuned, by grid search on 3 **tune** clips (EAR ratio 0.6 → 0.5, stillness 0.01 → 0.02); results are reported on the other 6 **eval** clips (1161 person-seconds). I compare three systems: **naive** (eyes closed = asleep), **face-only** (state machine without body signals) and the **full** system.

4. Results

System (eval clips)	Asleep precision	Asleep recall	Asleep F1	Asleep IoU	Accuracy, 5 states
Naive	0.58	0.96	0.72	0.57	0.80
Face-only	0.82	0.96	0.88	0.79	0.77
Full system	**0.95**	**0.95**	**0.95**	**0.91**	**0.78**

- **Phone use** (looking down, 126 s): false “asleep” seconds — naive 117, face-only 48, full **0**.
- **Head on the desk**, face hidden (tune clip): face-only misses all 79 sleep seconds; the body path misses 1.
- **5 s eye closures:** all three reported as drowsy, none as asleep (naive: 15 false asleep seconds).
- **Dim light at 16 fps:** F1 0.95. **Two people:** both named correctly (face similarity 0.74–0.78 to the right photo vs 0.07–0.12 to the wrong one), F1 0.94, including across a tracker ID switch.
- **Speed:** 5–9 frames/s on a laptop CPU, i.e. 1–2.2× the video length, without a GPU.

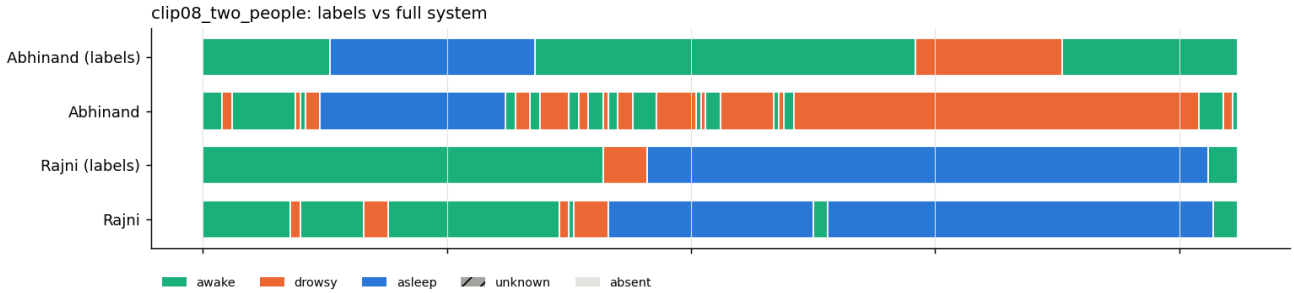


Figure 2. Two-person clip (0–212 s): hand labels above each prediction. Blue = asleep, orange = drowsy, green = awake.

How I got these numbers (honestly). The first evaluation gave eval F1 **0.61**. Looking at the eval failures found three bugs, each now fixed and covered by a test: (1) the per-event IoU metric mixed clips; (2) in the phone clip I looked down from the first second, so “head down” was learned as normal and 74 s were called asleep (fix: calibrate only on camera-facing frames, plus a 20° absolute head-down limit); (3) a 0.4 s overlap at a tracker ID switch left half of Rajni's sleep unnamed (fix: allow 1 s at a handover). I also re-checked four disputed label spans against the video and corrected them (one “sleep” clip was really phone use). Both steps used eval clips, so **0.95 is optimistic**; the tune-split F1 (0.96) did not change.

5. Failure cases and limitations

- **Phone vs dozing off:** head tilt measured +15° to +31° on the phone and about +18° when dozing with the chin dropped. The 20° limit separates them on my clips, but only by a few degrees. A phone detector (YOLO's “cell phone” class near the hands) would be a more robust signal.
- **Too much “drowsy”:** 225 of 794 awake seconds were called drowsy, mostly the nodding rule firing on normal head movement (Figure 2, Abhinand 120–205 s). It does not affect the asleep scores. Fix: require repeated dips, or look at nod frequency.
- **Small, acted data:** two people, one room, one camera, fake sleep. The body-only path (facing away) is tested on only one clip, which was in the tune set. Real sleep looks different (slow slumping, deep breathing).
- **Offline design:** once sleep is confirmed, its first 10 s are marked asleep too. A live webcam version would report onset 10 s late, but could reuse every part as is.

Next steps: more people, rooms and real naps; a phone detector; a better nodding rule; a live mode.